

Contract Administration AI

Automated Daily Work Report reconciliation for Ontario MTO construction projects



Local LLM pipeline: PDF → structured extraction → deterministic reconciliation → audit trail → review interface



Ollama / Llama 3.2



Pydantic V2



SQLite



HTML demo

Namyun Kim

AI Product / Business Analyst | Regulated Workflow Automation

This case study demonstrates

Regulated workflow analysis

Turning form-heavy construction administration into structured requirements.

AI product delivery

Local LLM extraction, validation, deterministic reconciliation, and human review.

Business value framing

Measured demo results separated from estimated ROI and future roadmap.

Namyun Kim

namyun.kim@ue-germany.de | [linkedin.com/in/namyun-kim-730626182](https://www.linkedin.com/in/namyun-kim-730626182) | github.com/Nami3777

I build AI workflows that make complex real-world processes reviewable, measurable, and safer to automate.

PROBLEM

Manual DWR reconciliation creates audit and payment friction

Current Pain Points

15–20 hrs

manual comparison per week

400+ hrs

manual verification per project

5–7%

error rate in variance detection

±5%

MTO variance threshold enforcement

Workflow & Opportunity



Current workflow

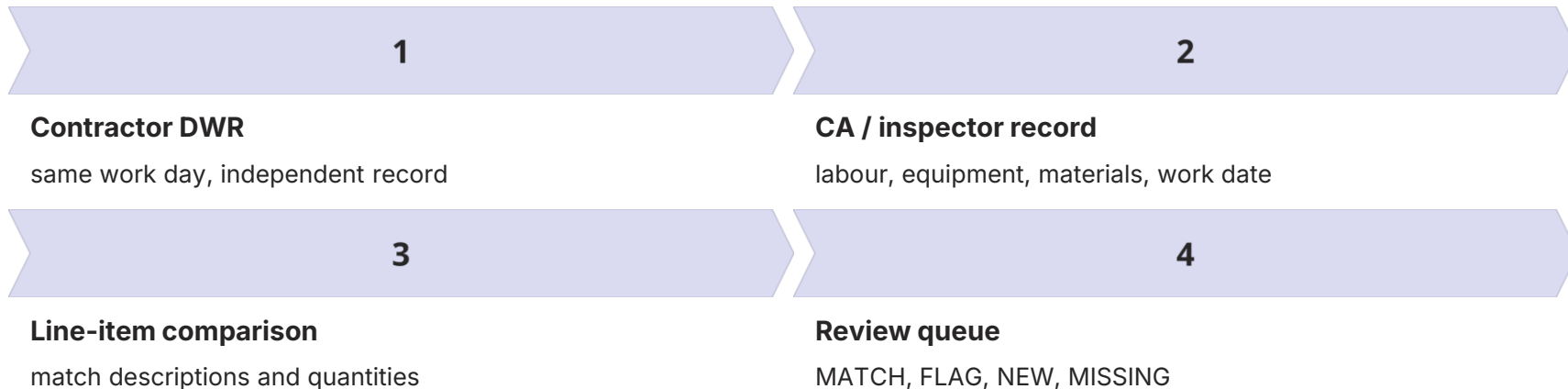
Contractor submits DWR → CA/inspector record exists → administrator compares line items manually → payment or dispute follow-up waits on reconciliation.



Product opportunity

Move administrators from data entry to exception handling: extract structured data, calculate variance deterministically, and route only review-worthy items to humans.

The workflow reconciles contractor claims against CA records



❗ The LLM structures unstructured DWR text. Python performs arithmetic and threshold decisions. Humans remain accountable for final approval.

The architecture separates extraction from financial decisions

01

Ingestion

PDF input converted into text / markdown through a Docling-first path with PyMuPDF fallback.

03

Reconciliation

Pure Python compares CA vs contractor line items and applies the $\pm 5\%$ threshold.

Contract Administration AI Case Study

04

02

Extraction

Ollama + Llama 3.2 extracts structured fields. Pydantic V2 enforces schema validation.

04

Storage & Demo

SQLite audit trail, console reporting, and interactive HTML interface.

Local inference is a governance choice, not a novelty

Selected Path

Ollama + Llama 3.2

Runs locally for sensitive construction records. No external cloud model is used in the implemented extraction path.

Risk Control

 Schema validation catches malformed outputs

 Retry logic handles extraction failures

 Deterministic code calculates variance

 Human review handles exceptions

Validation turns extracted text into a reviewable workflow

Type

quantities, dates, units,
categories

Range

impossible values and outliers

Logic

cross-field consistency and
status rules

Human review

final approval for exceptions

☐ Trust rule

AI identifies and structures evidence.
The system shows its work.
The administrator decides.

RESULTS

Measured demo results are separated from business estimates

Demo Results

20 min → 5 sec

Demo time per DWR pair

98.5%

Extraction accuracy with validation

100%

Deterministic variance calculation

Manual Review

Flags verified in Demo set

Business Impact Estimate

- ❑ Reduced administrator review time, faster discrepancy detection, fewer payment delays, and stronger audit readiness. ROI assumptions should be presented as estimates, not measured prototype results.

A product analyst view: requirements, acceptance criteria, and states

User Story

As a Contract Administrator, I need contractor and CA DWR line items compared automatically so I can review only discrepancies requiring judgment.

Acceptance Criteria

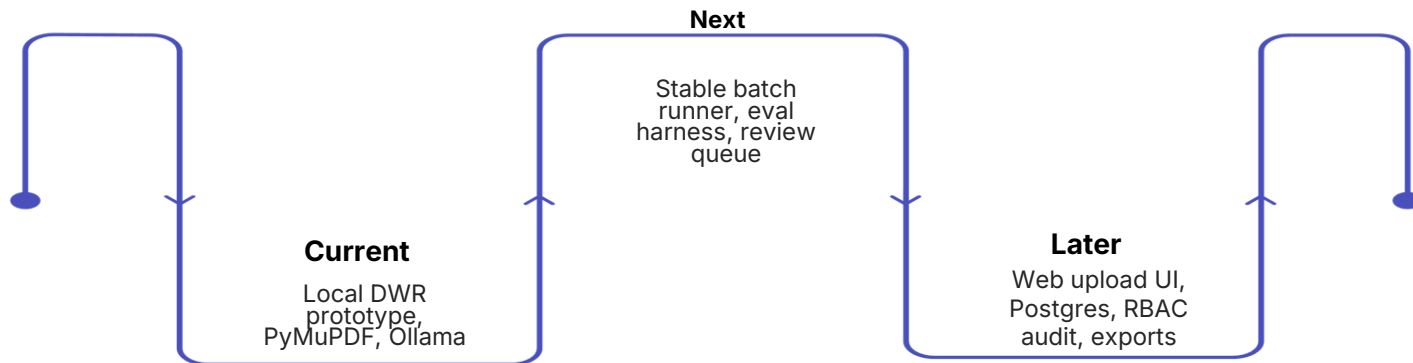
- Extract labour/equipment/materials
- Apply $\pm 5\%$ threshold
- Assign MATCH / FLAG / NEW / MISSING
- Persist source and timestamp

Metrics

- Time per DWR pair
- Extraction accuracy
- Flag review rate
- False positive rate

❑ The project is not just a script. It is a structured workflow with requirements, states, acceptance logic, metrics, and human review.

Roadmap: from portfolio demo to production workflow



Each phase builds on the last, moving from a local prototype to a robust, reviewable production workflow.

